

UCLA
Department of Statistics

STAT 200B: THEORETICAL STATISTICS

Problem Set 3
Winter 2020

Issued: Saturday, Jan 25

Due: Saturday, Feb 8

Full credit: 110 points.

Problem 3.1 (30 points)

Keener problem 4.23.

23. For $\theta > 0$, let

$$\Delta_\theta = \{(x, y) \in \mathbb{R}^2 : x > 0, y > 0, x + y < \theta\},$$

the interior of a triangle. Let $(X_1, Y_1), \dots, (X_n, Y_n)$ be i.i.d. from the uniform distribution on Δ_θ , so their common density is $21_{\Delta_\theta}/\theta^2$.

- a) Find a complete sufficient statistic T .
- b) Find the UMVU estimators of θ and $\cos \theta$.

Problem 3.2 (15 points)

Keener problem 4.21

21. Let $X_1, \dots, X_n$ be i.i.d. from an exponential distribution with density

$$f_\theta(x) = \begin{cases} \theta e^{-\theta x}, & x > 0; \\ 0, & \text{otherwise.} \end{cases}$$

Find UMVU estimators for θ and θ^2 .

Problem 3.3 (30 points)

Suppose that (X_i, Y_i) , $i = 1, \dots, n$ are i.i.d. samples from the two-dimensional normal distribution

$$\begin{bmatrix} X \\ Y \end{bmatrix} \sim N\left(\begin{bmatrix} 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 1 & \theta \\ \theta & 1 \end{bmatrix}\right).$$

with $\theta \in \Omega = (-1, 1)$.

- (a) Find a two-dimensional minimal sufficient statistic.

Hint: Read about curved exponential families.

- (b) Prove that the minimal sufficient statistic found in (a) is not complete.
- (c) Prove that $Z_1 = \sum_{i=1}^n X_i^2$ and $Z_2 = \sum_{i=1}^n Y_i^2$ are both ancillary, but that (Z_1, Z_2) is not ancillary.

Problem 3.4 (20 points)

Let $X_1, X_2, \dots, X_n$ be i.i.d. draws from $N(\sigma, \sigma^2)$ where $\sigma > 0$ is the unknown parameter.

- (a) Show that $T = (\sum_i X_i, \sum_i X_i^2)$ is minimal sufficient.
- (b) Find an unbiased estimator of σ based on $\sum_i X_i$.
- (c) Find an unbiased estimator of σ based on $\sum_i X_i^2$.
- (d) Is T from part (a) complete? Justify your answer. *Hint: use parts (b) and (c).*

Problem 3.5 (20 points)

Suppose that $X_1, \dots, X_n$ are i.i.d. samples of the discrete RV with mass function

$$f(x; \theta) = \begin{cases} \theta & \text{if } x = -1 \\ (1 - \theta)^2 \theta^x & \text{for } x = 0, 1, 2, \dots \end{cases}$$

- (a) Show that $(X_1, \dots, X_n)$ can be written as a 2-dimensional exponential family in terms of $T_1(x) = \sum_{i=1}^n \mathbb{I}[x_i = -1]$ and $T_2(x) = \sum_{i=1}^n x_i \mathbb{I}[x_i \geq 0]$.
- (b) Is (T_1, T_2) complete? How does this relate our result from class on completeness in exponential families?

Problem 3.6 (30 points)

Let $X = (X_1, X_2, \dots, X_n)$ be collection of independent random variables. Then,

$$\text{var} \left(\sum_{i=1}^n X_i \right) = \sum_{i=1}^n \text{var}(X_i). \quad (1)$$

The goal is to provide a partial extension of this result to a general function $Z := f(X)$, assuming Z is square-integrable (i.e., $\mathbb{E}Z^2 < \infty$).

For any random variable U , let $\mathbb{E}_i[U] = \mathbb{E}[U \mid X_1, X_2, \dots, X_i]$ and $\mathbb{E}^{(i)}[U] := \mathbb{E}[U \mid X^{(i)}]$ where $X^{(i)} := (X_1, X_2, \dots, X_{i-1}, X_{i+1}, \dots, X_n)$. We will prove the *Efron–Stein inequality*:

$$\text{var}(Z) \leq \sum_{i=1}^n \mathbb{E}[(Z - \mathbb{E}^{(i)}Z)^2]. \quad (2)$$

Let $\Delta_i := \mathbb{E}_i[Z] - \mathbb{E}_{i-1}[Z]$ (Doob's martingale difference).

- (a) Show that for $j > i$, $\mathbb{E}_i \Delta_j = 0$. Conclude that $\text{var}(Z) = \sum_{i=1}^n \mathbb{E}[\Delta_i^2]$.
- (b) Show that $\mathbb{E}_i[\mathbb{E}^{(i)}Z] = \mathbb{E}_{i-1}Z$.
- (c) Prove (2).

For two random variables U and V , we say that U is measurable w.r.t. to V (or the σ -field generated by V) if U can be written as a measurable function of V .

- (d) Show that

$$\text{var}(Z) \leq \sum_{i=1}^n \mathbb{E}[(Z - Z_i)^2]. \quad (3)$$

for any sequence $Z_i, i = 1, \dots, n$ of square-integrable random variables where each Z_i is measurable w.r.t. $X^{(i)}$.